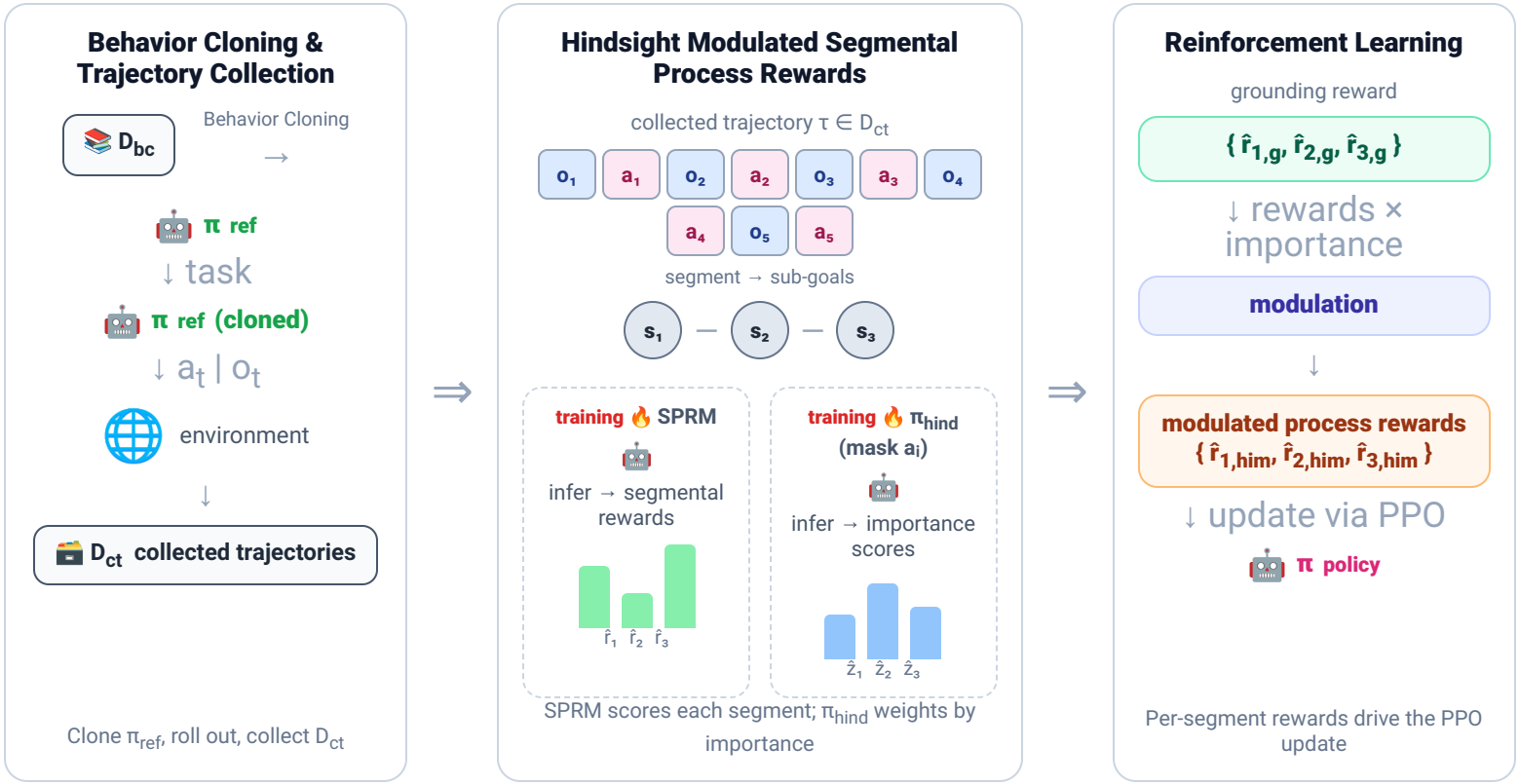


SeRPO: Segment-level Rubric Process Rewards for AppWorld Agents



Why segment-level process rewards matter

LEARNS WHERE OUTCOME RL CAN'T

On **fail-only** trajectories every rollout shares the same outcome, so outcome-reward baselines (Vanilla GRPO, GiGPO) get equal rewards $\rightarrow$ group variance 0 $\rightarrow$ **gradient = 0** (no learning). SeRPO's **per-segment** rewards still vary within a trajectory $\rightarrow$ **gradient $\neq$ 0**, so the policy keeps improving even when all trajectories failed.

LOCATES THE FAILURE POINT

A single outcome scalar says only *pass* / *fail*. Per-segment rewards $\{\hat{r}_1, \hat{r}_2, \hat{r}_3\}$ attribute credit along the trajectory, so a low-scoring segment **pinpoints which step caused the failure** (e.g. a premature action with missing API exploration) — turning an opaque rollout into a diagnosable, interpretable signal.

	Segment 1	Segment 2	Segment 3	Segment 4	Notes
Vanilla GRPO / GiGPO outcome only	X	X	X	X	all equal $\rightarrow \Delta=0$ X no gradient
SeRPO per segment	5	4	1	4	scores vary $\rightarrow \Delta \neq 0$ ✓ still learns

Same failed trajectory (outcome = fail). Segment scores reward the parts that worked and punish the failure point — so learning continues where outcome-RL gets nothing.